

Multiple-Level Distillation for Video Fine-Grained Accident Detection

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Abstract—Accident detection in surveillance or dashcam videos is a common task in the field of traffic accident analysis by using videos. However, as accidents occur sparsely and randomly in the real world, the data records are more scarce than the training data for standard detection tasks such as object detection or instance detection. Moreover, the limited and diverse accident data makes it more difficult to model the accident pattern for fine-grained accident detection tasks analyzing the accident in detail. Extra prior information should be introduced in the tasks such as the common vision feature which could offer relatively effective information for many vision tasks. The big model could generate the common vision feature by training on abundant data and consuming a lot of computing time and resources. Even though the accident video data is special, the big model could also extract common vision features. Thus, in this paper, we propose to apply knowledge distillation to fine-grained accident detection which analyzes the spatial temporal existence and severity for solving the issues of complex computing (distillation to the small model) and keeping good performance under limited accident data. Knowledge distillation could offer extra general vision feature information from the pre-trained big model. Common knowledge distillation guides the student network to learn the same representations from the teacher network by logit mimicking or feature imitation. However, single-level distillation could only focus on one aspect of mimicking classification logit or deep features. Multiple tasks with different focuses are required for fine-grained accident detection, such as multiple accident classification, temporal-spatial accident region detection, and accident severity estimation. Thus in this paper, multiple-level distillation is proposed for the different modules to generate the unified video feature concerning all the tasks in fine-grained accident detection analysis. The various experimental results on a fine-grained accident detection dataset which provides more detailed annotations of accidents demonstrate that our method could effectively model the video feature for multiple tasks.

Index Terms—Video accident detection, fine-grained accident detection, knowledge distillation, multiple-level distillation.

I. INTRODUCTION

IN RECENT years, video traffic accident analysis has advanced dramatically [1], [2], [3], [4]. Because of the rapid expansion of the city scale and the rapid increase in vehicle number, the number of traffic accidents is unavoidably increasing. Traffic congestion and accidents have become the most difficult problems for traffic management departments. For smart city construction, much more precise accident detection is needed. Effective detection of traffic accidents can reduce the loss of property and casualties caused by delayed rescue. Therefore, it may not be feasible to use human resources to monitor and make decisions on every single screen. With the proliferation of camera equipment, it is now possible to use computer vision technology with monitoring data to automatically detect and analyze traffic accidents.

Even though traffic surveillance cameras and dashcam could extensively record the traffic situation, traffic accident records are still rare because the accidents occur sparsely and randomly in the real world. The traffic accidents are buried in a mass of normal traffic videos. Some researchers collect traffic accidents through surveillance cameras and dashcam for accident detection or accident risk prediction [2], [5]. Some datasets about traffic accidents are constructed for detecting accident occurrence [1], [2]. However, compared to the diversity of accident patterns, the quantity of datasets still appears slightly inadequate.

Traffic accidents are always regarded as an emergency state in traffic scenarios because of their suddenness and unpredictability. Accident detection is defined as a subtask of abnormal detection. However, the analysis of accidents should be distinguished from the analysis of human anomalies because traffic and crowd are two scenarios with large differences.

Common traffic accident analysis usually focuses on the occurrence time of the accident, making the accident detection as the 0/1 binary classification [1], [5]. However, accidents occur in complex situations in the real world, so more comprehensive analyses about traffic accidents are needed. Therefore, we introduce multiple tasks for fine-grained accident detection to analyze the traffic accident in more detail, considering the multiple accident classification, temporal-spatial accident region detection, and accident severity estimation.

The big model could generate the common vision feature by training on abundant data and consuming plenty of computing

Manuscript received 28 July 2023; revised 27 September 2023; accepted 27 November 2023. Date of publication 1 December 2023; date of current version 6 June 2024. This work was supported in part by the China Postdoctoral Science Foundation Funded Project under Grant 2021M691683; in part by the Major Project of PCL under Grant PCL2023A08; in part by the National Natural Science Foundation under Grant U20B2052, Grant U20A20184, Grant U21B2038, Grant 62236008, and Grant 61931008; and in part by the Fundamental Research Funds for the Central Universities under Grant E2ET1104. This article was recommended by Associate Editor J.-M. Guo. (Corresponding author: Xinfeng Zhang.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2023.3338743>.

Digital Object Identifier 10.1109/TCSVT.2023.3338743

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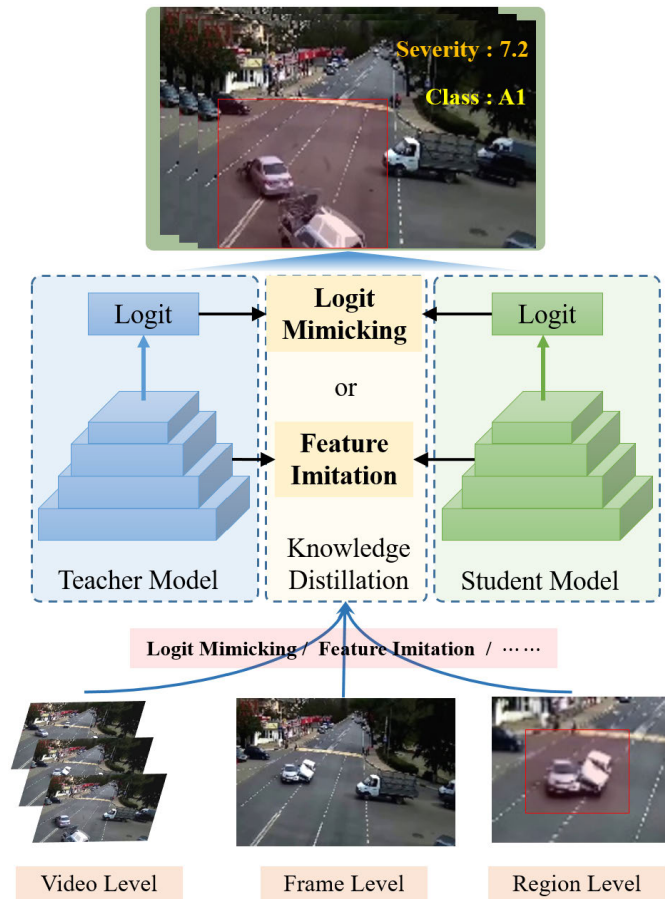


Fig. 1. The two kinds of knowledge distillation methods consider different targets of vision knowledge: feature imitation and logit mimicking. For video fine-grained accident detection, a multiple-level distillation strategy should be designed with thorough consideration as various tasks (i.e. multiple accident classification, temporal-spatial accident region detection, and accident severity estimation) focus on different levels of information using logit mimicking, feature imitation, and other strategies.

time and resources which could offer a relatively effective expression for many vision tasks with the same feature. The video data records could describe the traffic accident in much more detail, offering particular information about the traffic accident, but the models using video data require a lot of computation at the same time. More traffic accident record data are indispensable especially for fine-grained accident detection, at the same time, how to use the existing accident data and the data or methods of other computer vision tasks also need to be studied.

To address such issues, we propose to apply the knowledge distillation for fine-grained accident detection in this paper, making the model light weighted with less computing resources and achieving good performance with limited accident data.

The knowledge distillation framework could make the student network learn from the feature pattern of the teacher model, extracting the common vision features or more distinguishing features. The two kinds of knowledge distillation methods focus on the different targets of the vision knowledge: feature imitation and logit mimicking, as shown in the upper part of the Figure. 1. Feature imitation makes the students' features match the features of the teacher

in some stages as much as possible. Conversely, the logit mimicking methods aim to make the distribution of the student's final logit results match the results generated by the teacher model. The two kinds of knowledge distillation have different priorities. Meanwhile, effective knowledge for various tasks has the corresponding characteristics such as the classification task needing global information and the object detection task focusing on the local information.

For fine-grained accident detection, the accident videos are analyzed at various levels as shown in Figure. 1. As we introduce knowledge distillation into fine-grained accident detection for various tasks, the distillation should consider multiple emphasis points to extract more comprehensive and overall video visual features. Thus, the different knowledge distillation strategies should have the ability to improvise for the various analysis levels. In this paper, we propose multiple-level distillation for the different modules to concern all the tasks in fine-grained accident detection analysis, especially considering the difference between the RGB feature distribution and the optical flow feature distribution.

The main contributions of this paper are summarized in three folds below:

- We introduce the knowledge distillation into fine-grained accident detection for various tasks including multiple accident classification, temporal-spatial accident region detection, and accident severity estimation.
- We propose multiple-level distillation for the different modules to concern all the various tasks in fine-grained accident detection.
- Extensive ablation studies and comparative experiments on different fine-grained accident detection tasks demonstrate the effectiveness of our methods.

II. RELATED WORK

A. Anomaly Detection

Traffic accidents are always regarded as a subcategory of abnormal detection. The accident investigation must continue to distinguish between normal and abnormal conditions. Anomaly detection has long been studied, with the goal of locating abnormal events in a video sequence [6], [7]. Because modeling abnormal events is difficult, the abnormal status is defined as the most different from normal behaviors, such as bikers, small carts, and cars among people in UCSD Ped [8] and ShanghaiTech [9], wrong direction and no payment in Subway [10], running or throwing in public [11], [12], jay-walking or car illegally parked on street [13], and crime in real-world such as UCF Crime [14]. However, specific traffic accident detection could be more precisely defined than general anomaly detection.

The anomaly definition is more akin to an unsupervised learning framework, with only normal data used for training and both normal and abnormal data used for validation. Because normal status is more predictable than abnormal status in some temporal dependency contexts, [9] uses a video sequence with consecutive frames to generate a future frame as close to the ground truth as possible by adding intensity, gradient, and motion constraints. The constraints

ensure consistency in RGB space and reduce the difference in optical flow. Jardim et al. [15] use a low-rank sparse representation of target videos with a corresponding similar decomposition performed on an anomaly-free reference video. Reference [3] introduces a self-trained deep ordinal regression model to learn a ranking model for unsupervised video anomaly detection that optimizes the anomaly score. The attention-driven loss [16] is proposed to address the problem of foreground-background imbalance in anomaly detection. A novel distance-based anomaly detection method is proposed to take advantage of all the available normal data efficiently and flexibly [17], using locality-sensitive hashing (LSH) to map the samples.

Meanwhile, according to Zhong et al. [18], anomaly detection is a weakly supervised learning problem solved by noisy label learning, where the labeling of some frames in abnormal videos is incorrect. They used more precise noise labels to train the Graph Convolutional Network so that the classification Network could be trained better with frame-level labels. Reference [19] introduces a high-order context encoding model that extracts semantic representations and dynamic variations, with a focus on temporally localizing anomalous segments within anomalous videos. A sequential updatable anomaly detection framework [20] is proposed considering the importance of updating the new knowledge from false alarm data, based on the RobbinMonro algorithm and a fast version of Mahalanobis distance. Semi-supervised learning is also used to detect anomalies, as demonstrated by Abati et al. [21], which jointly learn an autoencoder and a density estimation network with autoregressive network layers. There are also some supervised methods based on temporal annotated training videos, such as [22] introduce the object detector and multiple object tracking for generating anomaly vehicle trajectories.

B. Traffic Accident Analysis

In traffic scenarios, some common vision tasks are also important tasks for traffic analysis, such as trajectory learning [23], object segmentation [24], pedestrian detection [25], scene understanding [26], and so on.

As an accidental occurrences in traffic scenes, traffic accidents have a wide range of impacts. Much research on traffic accident analysis has been increasingly emphasized [4], [27], [28]. For traffic accident analysis in computer vision, two types of videos are typically used: the dashcam in the vehicle's interior and the stationary surveillance camera on the street.

The dashcam could provide the first view of the accident, displaying accident details from participants or witnesses in the area [1], [4]. Dashcams are useful for a variety of applications, including self-driving cars. Although it may be useful for understanding the scene situation and detecting or predicting the emergency, it may not be able to fully capture the overall perspective of the accident due to only displaying a limited view of the scene. The traffic accident analysis on dashcam videos mainly focus on detecting the time when the accident occurs or predicting the accident risk such as in [1] and [29]. An uncertainty-based accident

anticipation model with spatio-temporal relational learning is proposed using the graph convolution networks and recurrent networks for extracting relational features [4]. The bottom-up and top-down visual attention mechanisms are introduced in a dashcam observation environment in [27] for the explanation by attentive regions from the proposed stochastic multi-task agent.

Still, surveillance cameras are always placed at intersections or on either side of roads to help the official traffic management department observe and judge traffic conditions in a specific location [2], [30]. They can provide a much higher and wider view of the scene, recording the entire accident process. Reference [31] focuses on achieving high performance on congested roads by introducing track-compensated frame interpolation for tracking the vehicles and dealing with the massive number of CCTV cameras for crash estimation. A decision-tree enabled approach is proposed in [32] for extracting accident events from traffic cameras while accurately estimating the start and end time of the accidents.

C. Knowledge Distillation

Knowledge distillation could take the performance and generalization of a big model and apply it to a smaller network through distillation. The two major kinds of knowledge distillation methods focus on the different targets of vision knowledge: feature imitation and logit mimicking.

The feature imitation methods introduce the intermediate representations of the teacher's hidden layers to improve the training of the student model [33]. An evolutionary knowledge distillation approach is proposed to improve the transfer effectiveness of teacher knowledge, using an evolutionary teacher learned online and consistently transferring intermediate knowledge to supervise student network in [34]. The paraphraser and convolutional translator are proposed in [35] for extracting meaningful features in an unsupervised manner and learning the factors of the teacher network. To be specific, the attention map is used in [36] for significantly improving the performance of a student CNN network by forcing it to imitate the attention maps of the teacher network. The multiple intermediate hints in distillation are considered in [37], leading to better exploitation of the teacher model and a more distinctive discrepancy between the teacher model and student model. In [38] exclusivity-consistency regularized knowledge distillation is introduced for training the student network with unlabelled face data by the position-aware weight exclusivity and the hardness-aware feature consistency.

The logit mimicking methods mainly focus on the last output of the networks, mimicking the final prediction of the teacher model [39]. The compact and fast object detection networks are proposed in [40] with improved accuracy using knowledge distillation for simpler classification and region regression setups. Meanwhile, the effectiveness of the soft target for logit mimicking has also been verified [41]. For the various outputs of more specific tasks, fast pose distillation [42] is proposed for human pose estimation and localization distillation [43] for dense object detection. For alleviating the reliance of existing pedestrian detectors on

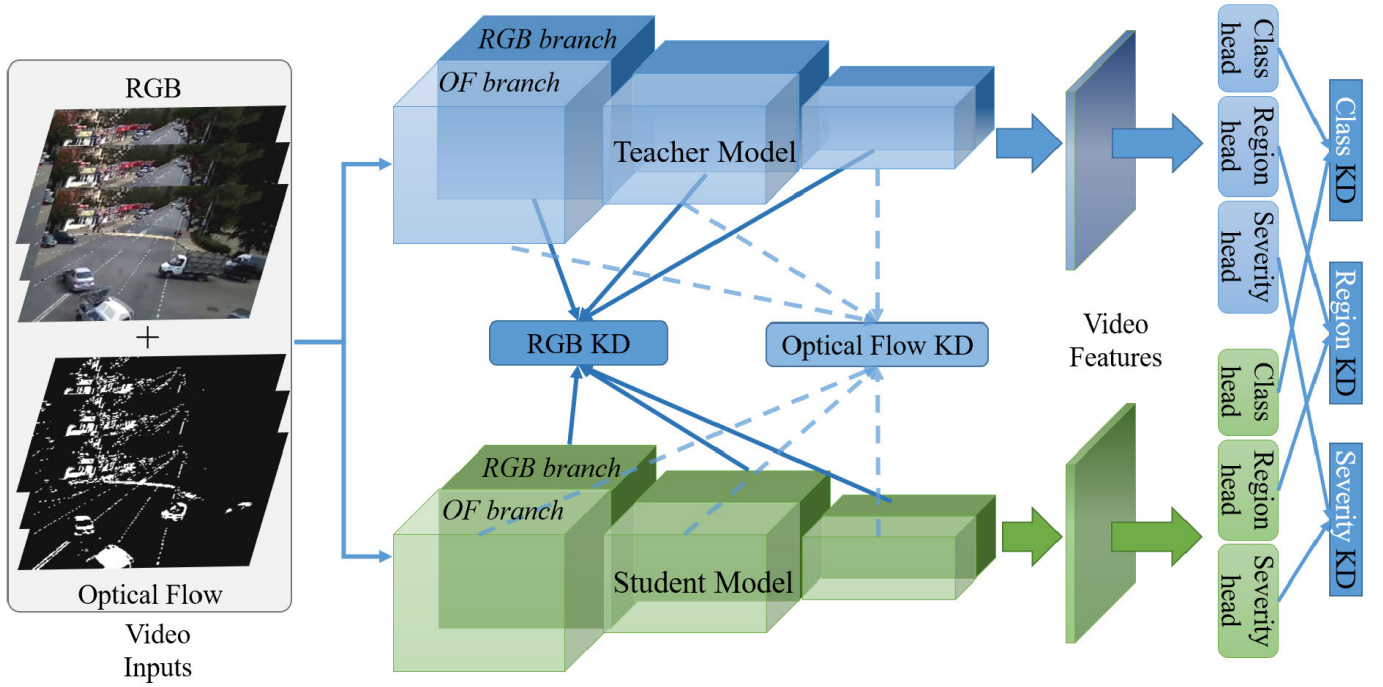


Fig. 2. The framework of the proposed model. Both the teacher model and student model contain the RGB branch and optical flow branch. We employ multiple-level knowledge distillation strategies including the RGB branch distillation, the optical flow branch distillation, and the multi-task heads distillation corresponding to the respective modules.

thermal images, the knowledge distillation framework is introduced to train a student detector, distilling the cross-modal representations guided by a well-trained detector [44].

Along with this research line, recent works have investigated many new techniques for propagating knowledge from the teacher network to the student network, such as mutual knowledge distillation [45], cross-resolution distillation [46], and multi-task learning [47]. For fine-grained accident detection in video, as we analyze the various tasks at the same time, the multiple-level knowledge distillation is considered for the respective requirement and characteristics of accident video.

III. METHOD

In this section, we provide a detailed introduction to multiple-level distillation for video fine-grained accident detection.

A. Architecture

For video fine-grained accident detection, the overall framework of the proposed multiple-level distillation is shown in Figure. 2.

As traffic accidents usually occur around the traffic participants such as the cars, trucks, motorbikes, or pedestrians on the road, we introduce the I3D network [48] for the teacher model which considers both the RGB information and optical flow information. The two branches framework could offer more effective video features considering both the appearance and motion information discussed in Section IV. The student network is designed to have two branches i.e. the RGB branch and the optical flow branch, which have the same blocks as the teacher model, and each block contains only one module.

This design aims to obtain a similar structure as the teacher model and keep the student network appropriately small.

The video feature combines the RGB feature and optical flow feature. As the video fine-grained accident detection contains various tasks, multi-task heads are used for inferring the individual output results from the unified video feature, i.e. the multiple accident classification head, temporal-spatial accident region detection head, and accident severity estimation head respectively.

To transfer the semantic knowledge from the teacher model, we employ multiple-level knowledge distillation strategies. For basic semantic knowledge, different distillation strategies for the RGB branch and optical flow branch. Meanwhile, for inferring semantic knowledge, each inferring head has an individual way of distillation.

B. Loss for Multi-Task Learning

We utilize the multiple losses for each video fine-grained accident detection task. The loss function of multi-task learning is formulated as

$$\mathcal{L}_{mtl} = \mathcal{L}_{cls} + \mathcal{L}_{bbox} + \mathcal{L}_{0/1} + \mathcal{L}_{sev} \quad (1)$$

The loss function contains three parts: the multiple accident classification loss $\mathcal{L}_{cls}$, the temporal-spatial accident region detection loss $\mathcal{L}_{bbox}$ with the position discrimination loss $\mathcal{L}_{0/1}$, and the accident severity estimation loss $\mathcal{L}_{sev}$.

The multiple accident classification loss $\mathcal{L}_{cls}$ is designed as a focal loss because of the imbalanced traffic accident data, which is generated by the results of multiple accident

classification and is defined as:

$$\mathcal{L}_{cls} = \sum_{i=0}^C -\alpha(1 - c_i \log \frac{1}{1 + e^{-p_i}})^{\gamma} c_i \log \frac{1}{1 + e^{-p_i}}, \quad (2)$$

where c_i is the video clip ground truth class label. The probability of each class is denoted by p_i . With the sigmoid cross-entropy loss, α is the factor influencing the shared weight of each class sample. γ is the modulating factor that reduces the weight of easy-to-classify data and causes the model to focus more on difficult-to-classify examples during training.

Similar to the one-stage object detection, the temporal-spatial accident region detection loss $\mathcal{L}_{bbox}$ consists of bounding boxes IoU loss $\mathcal{L}_{IoU}$ and center-ness loss $\mathcal{L}_{ctr}$, which are defined as:

$$\mathcal{L}_{bbox} = \mathcal{L}_{IoU} + \mathcal{L}_{ctr}, \quad (3)$$

$$\mathcal{L}_{IoU} = -g_1 \log(\text{IoU}(b_p, b_g) + \varepsilon), \quad (4)$$

$$\mathcal{L}_{ctr} = -g_r \log \frac{1}{1 + e^{-r}}, \quad (5)$$

where b_p and b_g are the predicted region and ground truth region of the accident, respectively. $\text{IoU}(b_p, b_g)$ is the Intersection over Union of the two bounding boxes. g_1 is the label of the positive class. ε is a minor item used to prevent unwanted zeros. $\mathcal{L}_{ctr}$ is the center-ness loss which the center-ness of a region at one single location is defined similarly as [49],

centerness = $\sqrt{\frac{\min(l,r)}{\max(l,r)} \times \frac{\min(t,b)}{\max(t,b)}}$, where the regression offset targets l, t, r, b are computed at the location by estimating the distance from the target center. g_r is the ground truth of center-ness at each place with the area bounding box and r is the projected center-ness.

The determination loss $\mathcal{L}_{0/1}$ is a soft-max cross-entropy loss similar to the binary classification, which is generated by the determination of whether an accident exists at one point and is defined as:

$$\mathcal{L}_{0/1} = \frac{1}{m^2} \sum -(g_0 \log q_0 + g_1 \log q_1), \quad (6)$$

where g_0 and g_1 are the values for the class's ground truth labels at each location of $m \times m$ position candidate queries. p_0 and p_1 are the probabilities of the negative or the positive class at the relevant place.

For the severity estimation loss $\mathcal{L}_{sev}$, we choose the regression loss for detection given by the smooth-L1 loss function, defined as:

$$\mathcal{L}_{sev} = \begin{cases} 0.5(s_p - s_g)^2 & \text{if } |s_p - s_g| < 1 \\ |s_p - s_g| - 0.5 & \text{otherwise} \end{cases} \quad (7)$$

where $|s_p - s_g|$ represents the absolute difference between the projected severity score s_p and the actual severity score s_g .

C. Loss for Multiple Level Knowledge Distillation

We utilize multiple distillation losses for different modules in our proposed framework. The loss function of multiple-level knowledge distillation is formulated as:

$$\mathcal{L}_{wd} = \mathcal{L}_{RGB_w} + \mathcal{L}_{OF_w} + \mathcal{L}_{head_w} \quad (8)$$

The loss function consists of three parts for different level knowledge distillation: the RGB branch distillation loss $\mathcal{L}_{RGB_w}$, the optical flow branch distillation loss $\mathcal{L}_{OF_w}$, and the multi-task heads distillation loss $\mathcal{L}_{head_w}$.

The RGB branch distillation loss makes the constraint between the RGB branch features of the teacher model and the student model. As the RGB information could provide basic guide information for classification and region detection, the student's RGB feature maps should imitate the teacher's feature more closely. Thus we introduce the Mean Square Error loss for RGB branch distillation, which is defined as:

$$\mathcal{L}_{RGB_w} = \frac{1}{n} \sum_{i=1}^n (f_{RGB_i}^T - f_{RGB_i}^S)^2 \quad (9)$$

where $f_{RGB_i}^T$ and $f_{RGB_i}^S$ are the teacher's feature and the student's feature of corresponding block i in the RGB branch.

The optical flow branch distillation loss makes the constraint between the optical flow branch features of the teacher model and the student model. For the optical flow branch, the features provide the motion information of the scene, indicating which region the focus should be concentrated on. Thus we make the distribution of the optical flow branch feature bringing into proximity between the teacher model and student model. The Kullback-Leibler divergence loss is introduced for optical flow branch distillation loss, which is defined as:

$$\mathcal{L}_{OF_w} = \frac{1}{n} \sum_{i=1}^n f_{OF_i}^T (\log(f_{OF_i}^T) - \log(f_{OF_i}^S)) \quad (10)$$

where $f_{OF_i}^T$ and $f_{OF_i}^S$ are the teacher's feature and the student's feature of corresponding block i in the optical flow branch.

For the multi-task heads distillation loss, as we make the basic video feature of the student model mimicking the teacher model, the multi-task heads should also consider the teacher model's heads. The corresponding distillation loss is designed for each head considering the characteristics of each task. The multi-task heads distillation loss is defined as:

$$\mathcal{L}_{head_w} = \mathcal{L}_{cls}^w + \mathcal{L}_{bbox}^w + \mathcal{L}_{0/1}^w + \mathcal{L}_{sev}^w \quad (11)$$

The classification distillation loss $\mathcal{L}_{cls}^w$, the region detection distillation loss $\mathcal{L}_{bbox}^w$ and $\mathcal{L}_{0/1}^w$ utilize the strategy of logit mimicking distillation.

$$\mathcal{L}_{cls}^w = \sum S(O_{cls}^T) (\log S(O_{cls}^T) - \log S(O_{cls}^S)) \quad (12)$$

$$\mathcal{L}_{bbox}^w = \sum S(O_{bbox}^T) (\log S(O_{bbox}^T) - \log S(O_{bbox}^S)) \quad (13)$$

$$\mathcal{L}_{0/1}^w = \sum S(O_{0/1}^T) (\log S(O_{0/1}^T) - \log S(O_{0/1}^S)) \quad (14)$$

where $S(\cdot)$ is the softmax function. $O_{cls}^T, O_{bbox}^T, O_{0/1}^T$ are the outputs of each head in the teacher model and $O_{cls}^S, O_{bbox}^S, O_{0/1}^S$ are the outputs of each head in the student model.

Meanwhile, the L2 loss is adopted for the severity distillation loss $\mathcal{L}_{sev}^w$,

$$\mathcal{L}_{sev}^w = \sum (O_{sev}^T - O_{sev}^S)^2 \quad (15)$$

After training the student model with the teacher model, only the student model and student heads are used for inferring the multiple accident classification, detecting accident regions, and estimating accident severity.

IV. EXPERIMENTS

In this section, we evaluate our proposed method on the Fine-grained Accident Detection dataset, which provides various annotation information on the accidents such as Accident categories, Start frame, End frame, Occurrence region, Weather, and Severity score, containing 1,603 raw videos and corresponding optical flow by Dense Inverse Search [50] of traffic accidents with various resolutions. The FAD dataset could offer video accident data covering various tasks, including multiple accident classification, temporal-spatial accident region detection, and accident severity estimation. The FAD dataset splits the accident data into half and a half for the training and testing set. We follow the setting and train both the teacher model and student model using the training set.

A. Evaluation Metrics

For multiple accident classification, we utilize the Average Precision (AP) for evaluating the correctness of identifying an accident category from a video. We not only consider the total categories but also the 0/1 classification (AP#2) and the specific accident classification (AP#26). In this way, precision, recall, and AP can be adopted to evaluate the methods for various classification criteria. We also introduce the Receiver Operating Characteristic (ROC) and Area Under Curve (AUC) for evaluating the methods.

For temporal-spatial accident region detection, we use mean Average Precision (mAP) at various Intersection over Union (IoU) threshold respectively, (IoU=0.33, 0.5, 0.66, 0.75 (AP₃₃, AP₅₀, AP₆₆, AP₇₅)) to assess the accuracy of the methods. If IoU is larger than a threshold, then the region prediction is positive to contain an accident, otherwise, it is negative.

For severity estimation, we use the Mean Squared Error (MSE) and Mean Absolute Error (MAE), which are the average of the squared and absolute errors respectively, and could better reflect the actual situation of the predicted value error.

$$MSE(y, f(x)) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 \quad (16)$$

$$MAE(y, f(x)) = \frac{1}{n} \sum_{i=1}^n |y_i - f(x_i)| \quad (17)$$

We also use the correlation coefficient for evaluating the severity estimation, i.e. Pearson Correlation Coefficient (PCC), which measures the linear relationship between the predicted value and ground truth.

$$PCC = \frac{\sum(f(x) - m_{f(x)})(y - m_y)}{\sqrt{\sum(f(x) - m_{f(x)})^2 \sum(y - m_y)^2}} \quad (18)$$

where $m_{f(x)}$ is the mean of the predicted values $f(x)$ and m_y is the mean of the ground truth y . Spearman's Correlation

Coefficient (SPC), which evaluates the relationship between the predicted value and ground truth using a monotonic function.

$$SPC = \frac{\sum(R(f(x_i)) - \overline{R(f(x))})(R(y_i) - \overline{R(y)})}{\sqrt{\sum(R(f(x_i)) - \overline{R(f(x))})^2 \sum(R(y_i) - \overline{R(y)})^2}} \quad (19)$$

where $\overline{R(f(x))}$ and $\overline{R(y)}$ are average ranks of the predicted values $f(x)$ and the ground truth y .

B. Implement Detail

We implement our model with TensorFlow. For the input, we evenly select 16 frames from each video clip as the initial N_i images and resize each frame to 224×224 . For knowledge distillation, the original I3D framework [48] is used for the base teacher model framework. Meanwhile, the pre-trained parameters trained on the UCF101 dataset [51] are introduced for the teacher model with better performance. The student network has the same blocks as the teacher model I3D, and each block contains only one module. The parameter number of the teacher model is 44M and the parameter number of the student model is 17M.

To compare with baseline models, we construct several models for training directly and compare our method with other knowledge distillation methods. The baselines are introduced as follows:

- **I3D_RGB**: The I3D framework [48] only with RGB information is directly trained on the FAD dataset.
- **I3D**: The I3D framework with two branches considering both RGB and optical flow information is directly trained on the FAD dataset.
- **UCF+HEAD**: I3D framework with frozen pre-trained parameters trained on the UCF101 dataset connects the multi-task heads.
- **I3D_small**: The student network of the I3D framework is directly trained on the FAD dataset.
- **I3D+UCF**: The teacher I3D framework with pre-trained parameters trained on the UCF101 dataset for initializing is trained on the FAD dataset.
- **FeatureKD**: Basic feature imitation knowledge distillation with the trained teacher model **I3D+UCF**.
- **LogitKD**: Basic logit mimicking knowledge distillation with the trained teacher model **I3D+UCF**.
- **LDKD**: The localization distillation for detection with the trained teacher model **I3D+UCF**.
- **EKD**: The evolutionary knowledge distillation method [34] for improving the transfer effectiveness of the trained teacher model **I3D+UCF**.
- **PKD**: The prior knowledge distillation [47] for multitask learning with the trained teacher model **I3D+UCF**.
- **RGB_OF_KD**: Our proposed method only uses the RGB branch distillation and the optical flow branch distillation loss without the multi-task heads distillation.

For training, the α is set 0.25, γ is set 2.0 and ε is set $1e-5$. We train both the baseline models and our model by setting the initial learning rate to 10^{-4} for 30 epochs,

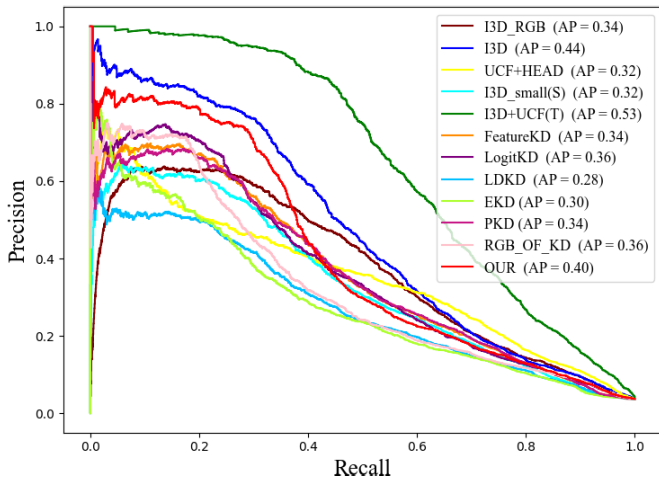


Fig. 3. Precision-Recall plot on the testing set of the FAD dataset. The legend presents the AP score of each method.

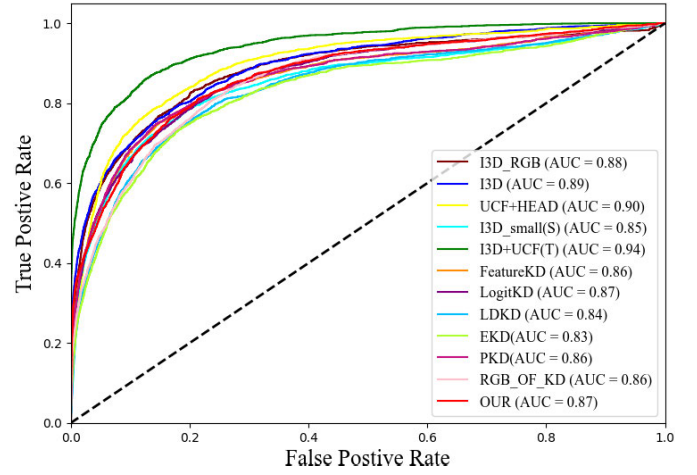


Fig. 4. ROC curve on the testing set of the FAD dataset. The legend presents the AUC score of each method.

10^{-5} for 30 epochs, 10^{-6} for 40 epochs as a total of 100 epochs with the batch size setting to 16 on an NVIDIA 3090 GPU.

C. Multiple Accident Classification

A quantitative comparison between our proposed method and the baseline models is shown in Figure. 3. We present the classical precision-recall curves on the testing set of the FAD dataset. The AP is computed on all the categories, which contain normal traffic videos and 26 categories of accident videos. From the figure, we could see that for directly training the models, I3D with pre-trained parameters trained for initializing on the UCF101 dataset achieves the best classification precision with 53%, because the pre-trained parameters bring more common video feature prior knowledge. Directly training student model I3D_small gets the worst result with 32% on AP slightly lower than only training the network heads with frozen pre-trained parameters. The student network could not generate the holistic video feature with few parameters for the accident classification. Only training the network heads has a similar problem. The comparison between the I3D_RGB and I3D demonstrates the advantage of the two branches framework for generating effective video features. For knowledge distillation with the trained teacher model I3D+UCF, our method achieves much better accuracy on AP than the other knowledge distillation baselines. Meanwhile, the LogitKD gets a better precision than the FeatureKD and localization distillation LDKD because of the final target mimicking. The distillation for multitask learning PKD gets good performance equilibrating the performance of different tasks. Our method without multi-task heads distillation RGB_OF_KD also gets good performance.

In consideration of the imbalance of samples in the categories in the proposed dataset, the receiver operating characteristic(ROC) curves on the testing set are shown in Figure. 4. The curves show that the result of our method is smooth and gets a higher AUC value, which means our model could be robust and accurate for video multiple accident classification. The I3D with pre-trained parameters also achieves the

TABLE I
COMPARISON WITH THE BASELINE METHODS ON VARIOUS AVERAGE PRECISION ON THE TESTING SET OF THE FAD DATASET. AP INDICATES THE AVERAGE PRECISION OVER ALL CLASSES. AP#26 INDICATES THE AVERAGE PRECISION OVER THE SPECIFIC ACCIDENT CLASSES. AP#2 INDICATES THE RESULTS OF BINARY 0/1 CLASSIFICATION REGARDING ALL THE ACCIDENTS AS THE POSITIVE SAMPLE

	AP	AP#26	AP#2
I3D_RGB	33.72%	18.28%	82.98%
I3D	43.83%	22.25%	87.81%
UCF+HEAD	32.09%	17.47%	64.93%
I3D_small(S)	31.56%	15.04%	71.05%
I3D+UCF(T)	53.42%	25.39%	98.19%
FeatureKD [33]	34.20%	15.71%	75.20%
LogitKD [39]	36.22%	15.14%	80.17%
LDKD [43]	27.91%	14.33%	62.91%
EKD [34]	30.12%	16.11%	66.74%
PKD [47]	33.76%	14.33%	74.84%
RGB_OF_KD	36.34%	16.47%	78.83%
OUR	40.42%	18.42%	82.69%

best AUC performance. Meanwhile, the FeatureKD, LogitKD, and RGB_OF_KD also achieve similar reliability performance.

We also compare our proposed method with the baselines on AP of only 26 accident categories and binary 0/1 classification, as shown in Table. I. For conventional binary 0/1 classification, we consider all kinds of traffic accident videos as the positive category and normal traffic videos as the negative ones. Both the proposed method and the baseline methods achieve more excellent performance on binary classification than multiple accident classification, especially the I3D+UCF gets 98.19% on AP#2. This result may occur because there are obvious differences in scene modes between normal traffic videos and traffic accident videos and the pre-trained parameters bring more basic vision knowledge. The original I3D significantly outperforms the small model I3D_small in terms of overall feature extraction. For knowledge distillation with the trained teacher model I3D+UCF, our method achieves much better accuracy on AP#2 than the other knowledge distillation baselines, which is comparable with the performance of directly training the original I3D. The LogitKD also shows excellent performance because of the global target mimicking. However,

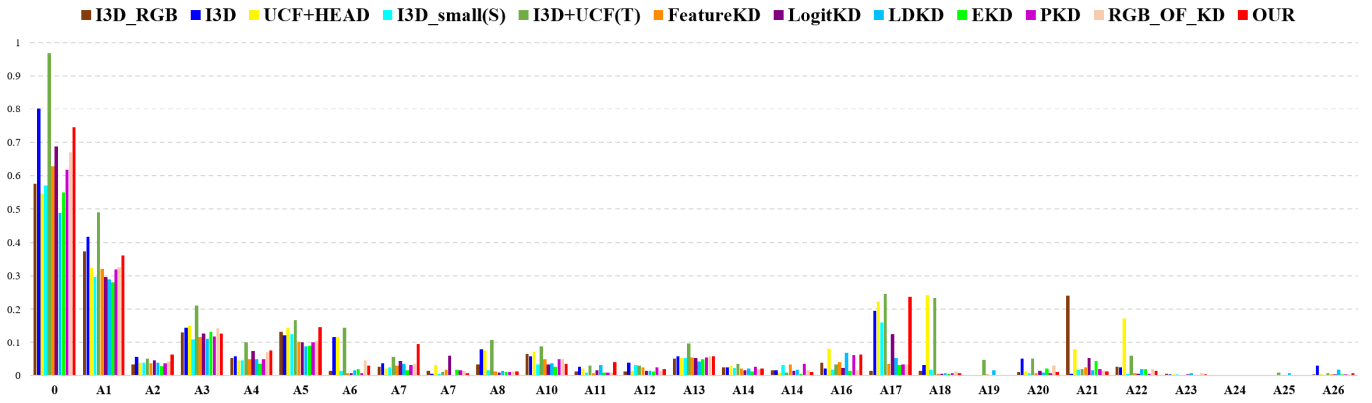


Fig. 5. Results of a quantitative comparison of the methods on each accident category.

TABLE II

COMPARISON WITH THE BASELINE METHODS ON THE AVERAGE PRECISION ACCORDING TO WEATHER ATTRIBUTE ON THE TESTING SET OF THE FAD DATASET

	Daylight	Night	Rainy/Snowy
I3D_RGB	32.96%	36.15%	37.64%
I3D	42.64%	46.19%	49.31%
UCF+HEAD	31.38%	33.39%	37.04%
I3D_small(S)	31.12%	33.57%	34.65%
I3D+UCF(T)	53.06%	56.73%	55.01%
FeatureKD [33]	33.27%	37.97%	36.60%
LogitKD [39]	36.01%	36.16%	40.36%
LDKD [43]	27.55%	27.97%	32.91%
EKD [34]	30.09%	29.68%	33.57%
PKD [47]	32.78%	37.87%	36.05%
RGB_OF_KD	35.74%	36.58%	41.74%
OUR	39.56%	43.89%	44.24%

TABLE III

COMPARISON WITH THE BASELINE METHODS ON VARIOUS AVERAGE PRECISION ON THE TESTING SET OF THE FAD DATASET. AP₇₅, AP₆₆, AP₅₀, AP₃₃ INDICATE AVERAGE PRECISION AT IoU=0.75, 0.66, 0.5 0.33, RESPECTIVELY

	AP ₇₅	AP ₆₆	AP ₅₀	AP ₃₃
I3D_RGB	0.24%	1.78%	10.59%	31.88%
I3D	0.49%	2.43%	15.94%	42.72%
UCF+HEAD	0.0%	0.81%	5.74%	17.64%
I3D_small(S)	2.75%	6.55%	19.74%	42.79%
I3D+UCF(T)	1.37%	5.02%	22.73%	45.80%
FeatureKD [33]	2.75%	6.39%	23.78%	46.68%
LogitKD [39]	2.10%	5.74%	20.95%	45.55%
LDKD [43]	2.67%	6.47%	21.27%	44.42%
EKD [34]	1.21%	3.24%	16.18%	38.19%
PKD [47]	2.51%	5.74%	22.25%	47.01%
RGB_OF_KD	1.70%	6.23%	22.73%	47.49%
OUR	2.27%	5.99%	21.76%	45.15%

the performances on the AP#26 are much worse which only evaluates the accuracy of the 26 accident categories. The I3D+UCF gets only 25.39% on AP#26 even the knowledge distillation methods with 15.71% 18.42%. Compared to other distillation methods, EKD performs well on the AP#26 using evolutionary distillation with mediocre AP#2. The reason for this decrease in performance is due to the variety of accidents. Even with the refined definition of the accident category, the fine-grained accident classification is quite challenging.

Figure. 5 displays the APs for each category. According to the graph, all of the models perform admirably on normal traffic videos of class '0'. The distribution of high and low performance on specific accident categories is roughly proportional to the data quantity distribution in the FAD dataset. This suggests that increasing the sample size of each fine-grained category could effectively learn the accident occurrence mode discriminatively. Meanwhile, prior vision knowledge brings much better performance. The results differ in some categories with small sample sizes, such as class A17 and A18. The reason for this could be that the participants in classes A17 and A18 include the 'Bus', which has a distinct appearance. The results of class A13 with more samples, on the other hand, are unsatisfactory due to various accidents.

To further investigate the models' performance in various scenarios, we evaluate the methods on various weather attributes in Table. II. According to the table, performance

is generally worse in *Daylight* than *Night* and *Rainy/Snowy*. The reason for this decrease in performance is that, while objects are clearer in daylight than at night, the background is more cluttered, and object association is more complex in daylight. Accidents are more easily distinguished in *Night* due to patterns influenced by light and darkness, whereas rain or snow makes the background more concise, resulting in a better performance in *Rainy/Snowy*. The I3D+UCF shows robustness in various scenarios with steady performance because of the common prior vision knowledge. The knowledge distillation methods could also get much better performance by using the learned knowledge of the teacher model than starting from scratch. Our method learns the global classification knowledge of the video better from the teacher model.

D. Temporal-Spatial Accident Region Detection

We compare the proposed method with the other methods on the testing set of the FAD dataset for the temporal-spatial accident region detection task. The results of AP at different IoU thresholds over which the region detection is considered positive are shown in Table III.

The table shows the result of temporal-spatial accident region detection is not satisfactory, as the best AP₇₅ of all methods is only 2.43%. Our proposed method achieves 21.76% at AP₅₀, which is comparable with the localization

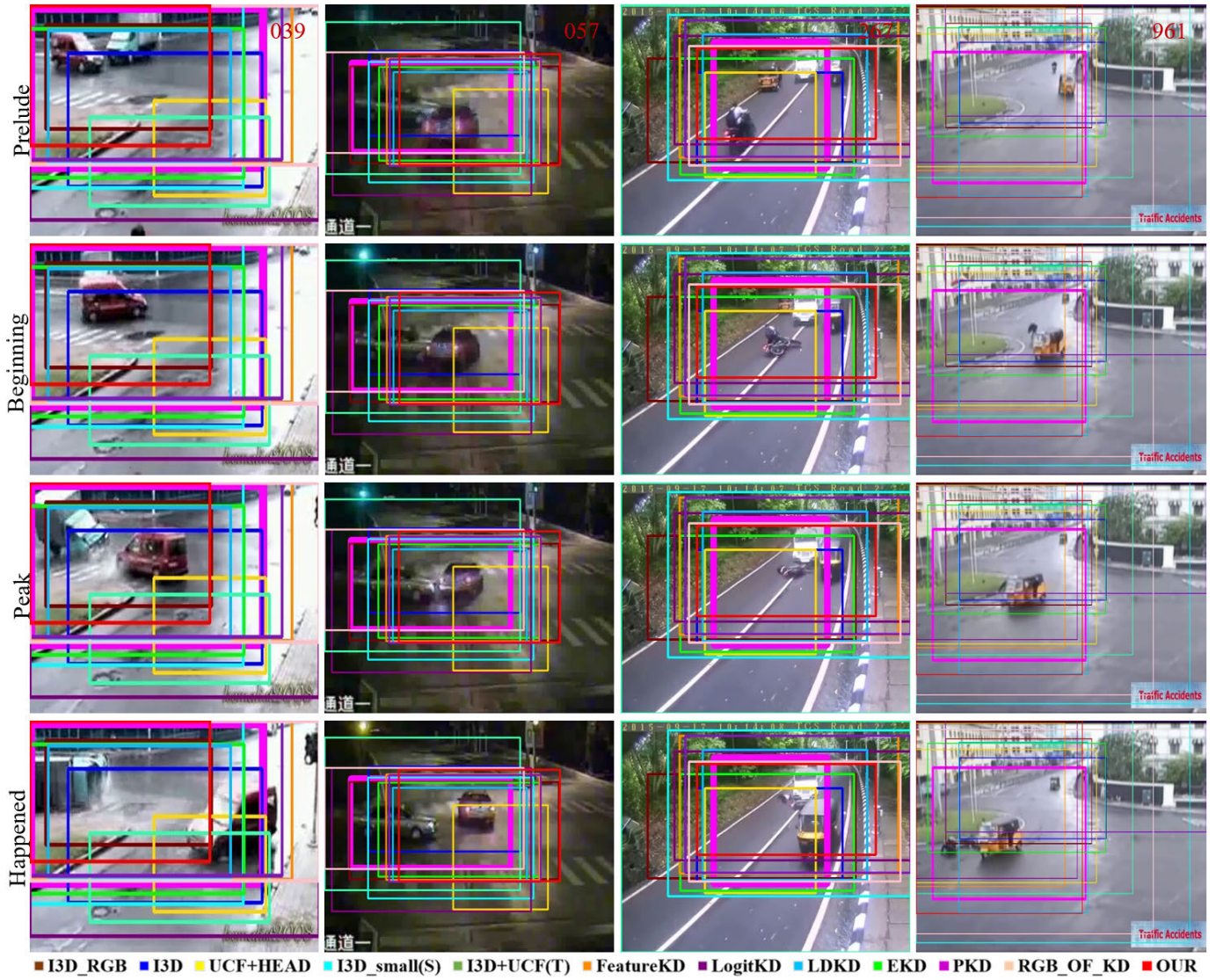


Fig. 6. Visualized comparisons of temporal-spatial accident region detection on the testing set of the FAD dataset. Each column shows the results of one video sequence, and the sequence number is shown at the top. Each row shows the prelude, beginning, peak, and happened state of the accident respectively. The magenta bold bounding boxes indicate the ground truth. The red ones indicate the results of our method. The other ones indicate the results of the other methods respectively.

distillation method LDKD. The feature imitation knowledge distillation method FeatureKD gets unanticipated best results 23.78% at AP_{50} . This is because the low-level feature is effective for region detection tasks. The FeatureKD keeps feature imitation learning the knowledge better from the teacher even though the small model focuses on normal patterns and reduces the generalization ability of anomaly patterns, producing more distinguishable representations from the teacher model as the small model I3D_small also achieves well performance. PKD achieves good detection results 22.25% at AP_{50} using distillation for equilibrating multitask. The localization distillation method LDKD gets better performance than the common logit mimicking LogitKD, while our multiple-level distillation method also achieves better detection results as we extract the video feature considering the respective characteristics of different branches.

We also evaluate the performance of the methods on various weather attributes, and the results are compared on the AP_{50} as shown in Table IV. The results of each model are

TABLE IV
COMPARISON WITH THE BASELINE METHODS ON AVERAGE PRECISION ACCORDING TO WEATHER ATTRIBUTE ON THE TESTING SET OF THE FAD DATASET. THE VALUES ARE THE AVERAGE PRECISION AT $IoU=0.5$

	Daylight	Night	Rainy/Snowy
I3D_RGB	11.15%	10.01%	9.09%
I3D	16.73%	16.02%	12.63%
UCF+HEAD	6.20%	3.90%	6.06%
I3D_small(S)	20.43%	18.61%	18.18%
I3D+UCF(T)	23.42%	21.65%	21.22%
FeatureKD [33]	23.54%	23.81%	24.74%
LogitKD [39]	23.30%	16.88%	16.16%
LDKD [43]	22.43%	17.75%	20.71%
EKD [34]	17.35%	13.42%	14.65%
PKD [47]	22.06%	21.21%	24.24%
RGB_OF_KD	23.05%	23.81%	20.20%
OUR	21.31%	22.08%	23.23%

quite different in weather attributes. This disparity may occur because the appearance feature is effective for detection with a clear scene in *Daylight* and lose the effect due to the

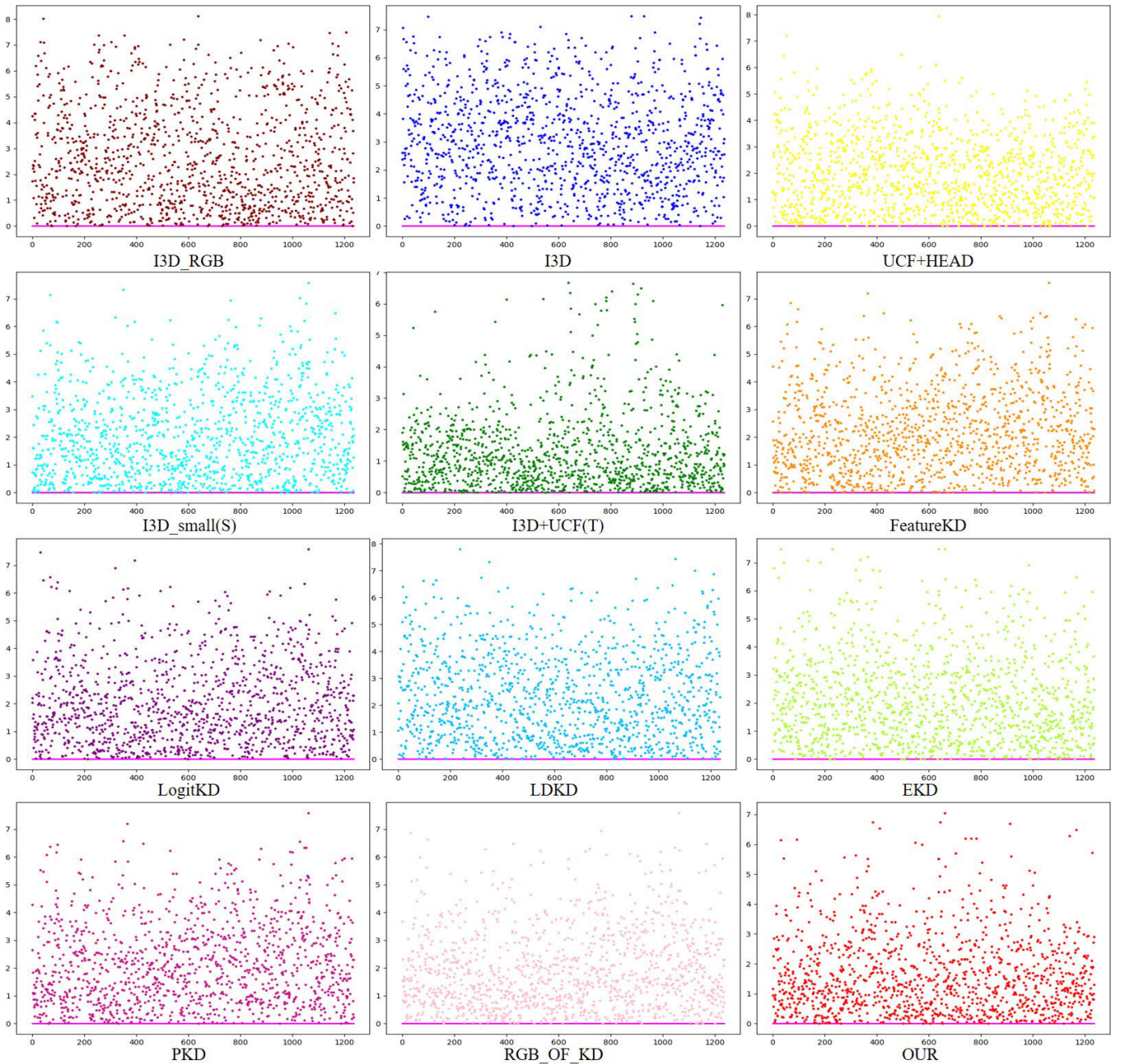


Fig. 7. Absolute Error of accident severity estimation on the testing set of the FAD dataset. The red ones indicate the results of our method. The other colors indicate the results of the baseline methods respectively. The abscissa represents the identity of the test video.

darkness in *Night* and the blur in *Rainy/Snowy*. The I3D+UCF shows the robustness of various weather attributes with steady performance in all *Daylight*, *Night*, and *Rainy/Snowy*. The proposed multiple-level distillation methods, multi-task distillation PKD and feature imitation FeatureKD could also get much better performance using the learned knowledge of the teacher model than the target logit mimicking LogitKD, EKD, and LDKD. Our method could learn the local region detection knowledge from the teacher model while keeping good global classification knowledge.

To demonstrate the capability of the proposed method, we visualize some temporal-spatial accident region detection results from the FAD dataset. Even though our method could

ideally learn to locate the occurrence region, as shown in Figure.6, the region's bounding box could not fit tightly to the real occurrence area. It implies that the temporal-spatial region detection of the accident in videos requires further investigation.

E. Accident Severity Estimation

We compare the proposed method with the baseline methods on the FAD dataset for the accident severity estimation task. The results of various metrics are shown in Table V.

Compared with the directly training methods, the I3D+UCF achieves much lower error performance on the MSE and

TABLE V

COMPARISON WITH THE BASELINE METHODS ON MSE, MAE, PCC, AND SPC ON THE TESTING SET OF THE FAD DATASET

	MSE	MAE	PCC	SPC
I3D_RGB	6.37	1.59	0.65	0.75
I3D	7.24	1.81	0.68	0.72
UCF+HEAD	3.64	1.22	0.82	0.78
I3D_small(S)	4.23	1.28	0.84	0.78
I3D+UCF(T)	1.91	0.79	0.88	0.82
FeatureKD [33]	4.86	1.41	0.86	0.79
LogitKD [39]	3.95	1.24	0.83	0.78
LDKD [43]	5.46	1.47	0.71	0.73
EKD [34]	4.45	1.30	0.74	0.73
PKD [47]	4.43	1.34	0.86	0.79
RGB_OF_KD	2.70	0.99	0.83	0.79
OUR	3.51	1.16	0.85	0.79

MAE, benefiting from the global generalization ability of video features and prior visual knowledge. For knowledge distillation with the trained teacher model I3D+UCF, the proposed multiple-level distillation method gets the best results with 2.70 at MSE and 0.99 at MAE. Particularly worth mentioning is that even without head distillation, our proposed method RGB_OF_KD also estimates accident severity well, getting better results than the other knowledge distillation methods. LogitKD gets better performance than LDKD, EKD, and PKD by focusing on the global target mimicking. The correlation coefficients between the ground truth severity score and the estimated result from the proposed method are comparable with the FeatureKD and LogitKD. It means that our methods achieve a relatively stable estimate of the accident severity while keeping less estimation error. While multiple people subjectively judge the severity scores of the accidents at various levels, our model achieves better ranked severity estimation from the SPC results.

To demonstrate the proposed method's capability, we visualize the absolute error results on the FAD dataset. The I3D+UCF estimates severity scores that are much closer to the ground truth, as shown in Figure. 7. Compared with other methods, our method learns the estimation ability of the teacher model better, resulting in a closer distribution of the severity estimation to the teacher model's results. Meanwhile, many severity estimation scores remain far from the truth, demonstrating how difficult the severity estimation task is. Our method could learn the global unified video feature for multiple accident classification and severity estimation while keeping local vision knowledge for temporal-spatial accident region detection.

F. Distilling by Using Less Data

We compare the proposed method with the baseline methods using half of the training set data for demonstrating the proposed method's robustness in the case of a small amount of data. The results of various metrics for the tasks are shown in Table VI.

We choose the main metrics for the respective tasks: AP for multiple accident classification, AP_{50} for temporal-spatial accident region detection, and MSE and PCC for accident severity estimation. According to the table, the performances of the methods descend distilling by using less training data

TABLE VI

COMPARISON WITH THE BASELINE METHODS ON AP, AP_{50} , MSE, AND PCC ON THE TESTING SET OF THE FAD DATASET

	AP	AP_{50}	MSE	PCC
FeatureKD [33]	23.19%	11.46%	5.24	0.71
LogitKD [39]	24.52%	10.44%	4.48	0.74
LDKD [43]	20.91%	11.57%	8.31	0.70
EKD [34]	24.43%	10.11%	4.87	0.71
PKD [47]	25.05%	12.78%	4.95	0.83
RGB_OF_KD	27.91%	14.89%	3.67	0.82
OUR	31.87%	16.10%	3.58	0.83

compared with using whole training data. Our method maintains a good performance in general with the least performance loss when using half training data. For multiple accident classification, more training data could offer more diverse accident patterns for robust discrimination ability. This situation is even more evident in temporal-spatial accident region detection. For accident severity estimation, our method achieves 3.58 on MSE with minimal prediction error overflow when using less data. Our method uses multiple-level knowledge distillation to learn both the global estimation knowledge and local temporal-spatial knowledge well from the teacher model.

V. CONCLUSION

In this paper, we introduce knowledge distillation into fine-grained accident detection for various tasks, including multiple accident classification, temporal-spatial accident region detection, and accident severity estimation. The multiple-level distillation is proposed for the different modules to concern all the various tasks in fine-grained accident detection. We utilize the RGB branch distillation, optical flow branch distillation, and multi-task heads distillation for learning both local and global video information from the teacher model. In addition, the extensive ablation studies and comparative experiments on different fine-grained accident detection tasks showed the effectiveness of our methods for generating the unified video feature on more challenging accident analysis tasks. In the future, we will consider extending temporal distillation using the image-based teacher model for generating effective video features.

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